

Martín Bernardo Bonilla

Computer Vision & Edge AI Engineer | Real-time On-device Pipelines
Argentina · Remote-ready

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Professional Summary

Computer Vision Engineer specialized in real-time on-device vision pipelines. I design and build end-to-end systems that combine facial landmark detection, geometric rendering, and video processing, with strong focus on performance, clean architecture, and modularity. Currently developing **Jax Glitchroom**, a local-first computer vision engine using MediaPipe and OpenCV that performs real-time 3D facial landmark tracking and visual effects at ~25 FPS on HD resolution. Solid background in building production-grade distributed systems with fault tolerance and rigorous testing.

Featured Projects

Jax Glitchroom — Real-Time Computer Vision & Biometric Telemetry Engine

2026 – Present · [Python](#) · [MediaPipe](#) · [OpenCV](#) · [NumPy](#) · [FFmpeg](#)

- Designed and implemented a local-first real-time video processing pipeline with 3D facial landmark extraction using MediaPipe Face Landmarker + XNNPACK.
- Achieved stable ~25 FPS on HD resolution (720×1280) with dynamic face detection and atomic JSON telemetry.
- Built multi-layer visual rendering (geometric and retro-futuristic effects) combined with audio processing using OpenCV and FFmpeg.
- Applied clean architecture principles (Domain, Application Ports, Infrastructure) for scalability and maintainability.
- Fully on-device execution with focus on memory stability and modular design for future features (liveness, kinetic analysis, depth estimation).

Biye — Distributed Payment E-Commerce Platform

2024 – Present · [Flutter](#) · [Dart](#) · [Node.js](#) · [Express](#) · [MongoDB](#) · [Redis](#) ·
[Mercado Pago](#) · [Docker](#)

- Built a fault-tolerant checkout system with webhook-first confirmation and polling reconciliation, ensuring zero lost transactions.
- Implemented idempotent event processing and deterministic order state management.
- Developed 41 automated tests covering critical edge cases and failure scenarios.
- Deployed with Docker on Railway and Vercel.

WoodStack — Personal Portfolio & Project Showcase

woodstack-portfolio.vercel.app · React · Vite · TailwindCSS · Vercel

- High-performance portfolio (96+ Lighthouse score) used as central hub for projects and CV download.
 - Modular component architecture with optimized asset loading and clean design system.
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Technical Skills

- **Computer Vision & AI:** MediaPipe, OpenCV, NumPy, real-time landmark detection, on-device inference
- **Languages:** Python, Dart, JavaScript/TypeScript
- **Backend:** Node.js, Express, MongoDB, Redis, Docker
- **Frontend:** Flutter, React
- **Others:** FFmpeg, Git, Clean Architecture, Testing